Fraudulent Claim Detection for Global Insure

Data-driven early classification of insurance claims using predictive modeling and feature engineering

Agenda

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Problem & Business Business Objective

Understanding the fraud challenge and strategic goals

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Data & Methodology

Data cleaning, feature feature engineering, engineering, and modeling pipeline

3

Key Findings

Exploratory data analysis insights from historical patterns

4

Model Results

Performance comparison and selected approach

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Business Impact

Recommendations and operational implications

Problem Statement

High Volume, High Stakes

Global Insure processes large claim volumes with a material share being being fraudulent, causing direct financial loss to the company

Late Detection Problem

Fraud is often discovered **after or during payout**, using slow and costly costly **manual review processes** that strain resources

Strategic Need

Flag likely fraud **early in the approval workflow** using historical claim and customer data to prevent losses

Business Objective

Primary Goal

Build a **binary classifier** that distinguishes fraud reported (Y) versus legitimate claims (N) using (N) using predictive modeling

Key Inputs

- Claim amounts and component breakdowns
- Customer and policy attributes
- Incident details and timing
- Historical fraud patterns

Success Metrics

Deploy the model to **score new claims before approval**, reducing paid fraudulent claims and claims and optimizing the investigation queue — without abandoning legitimate customers customers

Data Overview

~1K	~40	24.7%	700
Claims Analyzed	Raw Variables	Fraud Rate	Training Rows
Complete dataset of insurance claims with fraud labels	Policy, insured, incident, incident, vehicle, and claim claim components	Minority class in stratified stratified train/validation train/validation split	After 70/30 stratified stratified split (random_state=42)

Class imbalance confirmed: Fraud (Y) represents the minority class at approximately 24.7% in the analysis split. The split. The validation set contains 300 rows for unbiased performance evaluation.

Methodology Pipeline

Key EDA Insights

Patterns Revealed in Historical Data

Class Imbalance

Confirmed minority class distribution requiring balanced training approaches

Categorical Patterns

Incident types and police report availability show different fraud rates rates across categories

Component Relationships

Claim component splits (injury/property/vehicle) motivate motivate ratio-based features

These insights guided feature engineering decisions and highlighted valuable signals for fraud investigators.

Feature Engineering

Why Engineering Matters for Fraud Detection

Ratio Features

Injury_ratio, property_ratio, and vehicle_ratio capture **how a claim is split** across components components — often more informative than a single total amount

Calendar Features

Incident_month, incident_dow, policy_bind_year, and policy_age_days capture **seasonality seasonality and tenure effects** that influence fraud likelihood

Removed Variables

Dropped total_claim_amount to reduce multicollinearity; removed raw dates and noisy geography where EDA showed limited predictive value

Modeling Approach & Results

Logistic Regression

Interpretable model with RFECV RFECV selecting 23 features from from 102 dummy-encoded columns using ROC-AUC. Checked Checked p-values and VIF for multicollinearity. Decision threshold tuned to **0.57** (not default 0.5) for better fraud capture.

Random Forest

Strong on training data with tuned hyperparameters, but cross-validation suggested **overfitting** relative to validation validation performance on held-held-out data.

Fair Comparison

Both models trained on **balanced training set** (1,054 rows with RandomOverSampler), with performance reported on **raw raw validation distribution** (300 (300 rows, never oversampled). oversampled).

Validation: Logistic regression with tuned threshold and RFECV-selected features showed better generalization to unseen claims.

Validation Results

Metric	Logistic Regression	Random Forest
Accuracy	**0.8533**	0.8067
Recall (Sensitivity)	**0.7703**	**0.5405**
Specificity	0.8894	**0.8938**
Precision	**0.6951**	0.6250
F1-Score	**0.7308**	**0.5797**

Key Takeaway: Logistic Regression achieves higher recall and F1-score while maintaining strong specificity, making it the optimal choice for early fraud detection with reasonable false-alarm control.

Recommended Model for Deployment

Deploy Logistic Regression

Probability cutoff at **0.57** for production pilot

Why This Choice

False negatives costlier than false alarms; recall and F1 higher on validation

Caveat

If regulatory constraints dominate, revisit threshold or use scores for prioritization prioritization only

Question 1: Detecting Fraud Patterns

Patterns

Exploratory Data Analysis

- Examine class balance and distribution
- Compare univariate fraud vs non-fraud patterns
- Analyze category-level fraud rates
- Identify correlations and drivers

Feature Engineering

- Reflect business structure (component splits, tenure, timing) timing)
- Move beyond raw totals
- Use stratified train/validation splits
- Validate on imbalanced holdout data

Question 2: Predictive Features

01

Data-Driven Selection

RFECV reduced 102 encoded features to 23 strong strong predictors optimized via ROC-AUC under under cross-validation

02

Statistical Validation

Retained predictors supported by p-values with with acceptable VIF values showing controlled controlled multicollinearity

03

Thematic Clusters

Signals group around incident context, documentation patterns, claim composition ratios, ratios, and policy attributes

04

Complementary View

Random Forest feature importance provides additional perspective through tree-based importance scoring

Question 3: Predicting Fraud Likelihood

Yes, We Can Predict

Model pipeline outputs fraud probability from logistic regression with comparable scores from Random Forest

Forest

Validation Performance

On 300 validation claims, ~77% recall and ~85% accuracy ranks and flags true fraud cases effectively

Operational Workflow

Use scores for risk prioritization combined with rules and human investigation before final decisions

decisions

Question 4: Model Insights

Threshold Optimization

Default 0.5 cutoff is suboptimal; tuning tuning balances customer friction versus fraud capture

Target Investigations

Use top features to design checklists, checklists, document requirements, and requirements, and adjuster training training programs

Simpler Generalizes Better

Logistic regression beat Random Forest Forest on validation despite RF memorizing training data

Monitor Data Drift

Retrain models when claim mix, products, or fraud schemes change over over time

Business Implications

Financial Impact

Earlier detection reduces paid paid fraud and targets Special Special Investigation Unit effort on highest-risk claims claims

Operational Efficiency

Replace manual-only processes with score + rules + rules + human review, shortening cycle time for clear clear low-risk claims

Customer Trust

Tune cutoff and processes to to limit false positives that delay legitimate payouts and and frustrate customers

Risk & Compliance

Interpretable logistic regression supports documentation of claim escalation reasons with feature attributions

Analytics Maturity

Establishes repeatable data pipeline from cleaning to features to validation for ongoing model updates

Implementation Roadmap & Recommendations

Pilot Program

Deploy logistic regression scoring on new claims with cutoff 0.57; log outcomes for calibration and model monitoring

Workflow Integration

Integrate scores into tiered review workflow: auto-route low risk, fast-track high risk to fraud investigation team

KPI Alignment

Track recall, precision, F1-score, dollar-weighted loss prevented, and customer complaints from false flags

Model Refresh

Update quarterly or when fraud rate or features shift; maintain validation discipline with stratified or time-based splits

Future Extensions

Enrich with telemetrics, repair networks, and text from statements under privacy policy; current framework scales

Summary & Next Steps

What We Built

End-to-end fraud classification workflow on 1,000 claims with stratified 70/30 validation split

How We Handled Imbalance

Applied RandomOverSampler to training data only, preserving realistic validation conditions

Why Logistic Regression

Delivers best validation recall and F1-score versus tuned Random Forest; recommended as primary model primary model

Business Value

Directly supports early triage, investigation targeting, and measurable reduction in late fraud discovery discovery

Appendix Materials

ROC curves, precision-recall curves, confusion matrices, coefficient tables, RF feature importance, VIF analysis — all documented in notebook